

Quantised failure: trained reflection transports collapse onto exact group quotients

Technical report — draft v1

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Abstract

A recurrent layer whose state is moved by products of Householder reflections can be trained on the word problem of a finite group: read a sequence of group elements, report the product. When such a layer succeeds it is exact at any length, because it has learned a homomorphism. When it fails, it does not fail arbitrarily. Its accuracy lands on $1/|N|$ for a normal subgroup N — it has learned the quotient G/N , gets the coset right, and guesses inside it. We give the proven ceiling that makes the phenomenon measurable, the rung structure across several groups, a control at matched group order that separates the lattice from the order, a table-free diagnostic based on characters, and the tasks where a quotient is sufficient rather than a failure. We also report the same invariant appearing in an unrelated role: the normalised trace that identifies which quotient a transport learned is the quantity that governs how much information the same transport can bind and store.

1 Introduction

The word problem of a finite group is the cleanest state-tracking task there is: read a sequence of group elements, report the product. It has no ambiguity, no distributional shift, and a correct solution is exact at *every* length, because a model that solves it has represented a homomorphism. It is also the task on which the expressive limits of modern recurrent architectures are most sharply drawn — diagonal or element-wise transitions cannot represent a non-commutative group at all [11, 13], which is the observation that motivated negative eigenvalues [7] and products of Householder reflections [15] as transition parameterisations, and which connects the architecture question to a classical one about bounded-width computation [3].

This paper is about what such a model does when it *fails*. Train a reflection transport on the word problem of S_4 and one of two things happens. It reaches 1.000 and stays there as the evaluation length grows to eight times the training length. Or it settles on one of a small set of values — 0.250, 0.083, 0.042 — and stays there just as stably. Those are $1/4$, $1/12$ and $1/24$, and the normal subgroups of S_4 have orders 1, 4 (V_4), 12 (A_4) and 24. The correspondence is not a coincidence: a model that has learned the quotient G/N knows which coset of N the answer lies in and can do no better than guess inside it, so its accuracy is exactly $1/|N|$.

Failure, in other words, is quantised, and the allowed levels are fixed by the normal subgroup lattice of the group — an object the model never sees. We call these levels *rungs*. The contribution of this paper is to make that statement checkable rather than suggestive, and then to show that the algebra which explains it also predicts a second, apparently unrelated limitation of the same layer.

Contributions.

- **A proven ceiling** (Section 4) that separates commuting from non-commuting transports by a bound no training can move, computed exactly by enumeration, and a measurement of the reflection transport against it.
- **The lattice law** (Section 5): across Q_8 , S_4 and A_5 , twelve runs land on rungs of their own group with worst deviation 0.021; the kernel is identified independently from the model’s *errors* by coset consistency, and predicts the accuracy on 5/5 runs.

- **A same-order control** (Section 6) that separates the lattice from the group order, reported with its registered verdict — which one run falsified.
- **A table-free diagnostic** (Section 7): the character norm $\langle \chi, \chi \rangle$ identifies the learned quotient from traces alone. Over sixteen seeds it agrees with the independent error-based reading on all seven runs that are homomorphisms, and integrality of the irrep multiplicities selects exactly those seven runs without consulting the training objective.
- **Quotients as capability** (Section 9): when the task *is* a quotient, the same collapse is the correct answer, and the lattice law reappears one level down, on the rungs of $S_4/V_4 \cong S_3$ and $S_4/A_4 \cong \mathbb{Z}_2$.
- **The same invariant governs binding** (Section 8): the normalised trace that names the quotient is also what sets how much the transport can store, tying a state-tracking result to an associative-memory one.

What makes the law checkable. Three things, in increasing strength, each used below.

1. **Flatness.** A homomorphism is length-independent. Every claim below is scored far beyond the training length, and runs whose accuracy drifts with length are reported as such rather than scored against the law.
2. **Coset consistency.** If the model learned G/N , its *errors* must respect cosets: predicted and true element in the same coset of N , essentially always. A model that is merely bad has no reason to respect any coset.
3. **A proven ceiling.** For a commuting transport the product is order-independent, so the output can only be a function of the multiset of inputs; the best such function is computable by enumeration and is a hard bound.

Every derived quantity in this paper — the ceiling, the rung sets, the attainable character norms — is recomputed from the group definitions by a single script, which also re-reads the measured tables from the raw logs rather than trusting a transcription.

2 Related work

State tracking in linear recurrences. That a linear RNN with diagonal, positive transitions cannot track non-commutative state is by now well established [11, 13, 5], and two repairs are directly relevant here. Grazi et al. [7] show that allowing eigenvalues in $[-1, 1]$ — in our parameterisation, $\beta > 1$ — restores parity and related tasks; Siems et al. [15] generalise the DeltaNet transition [17, 16] to a product of n_h Householder reflections per token, which by the Cartan–Dieudonné theorem [2] can reach any orthogonal map for $n_h \geq d$. Our transport is that construction, and n_h is the knob whose two ends Section 8 argues are different jobs. Where that line of work measures *which tasks become solvable*, we measure *what the solution degrades to* when it does not, which turns out to carry more structure than a scalar accuracy.

Groups as a probe of computation. Using group word problems to characterise a computational model is classical: Barrington [3] shows that width-5 branching programs over S_5 capture NC^1 exactly, and non-solvability is the pivotal property. The same solvability boundary organises what shallow sequence models can learn [10, 5], and Krohn–Rhodes decomposition [9] gives the semigroup-theoretic version. We use groups differently: not to place the architecture in a complexity class, but because the normal subgroup lattice supplies a discrete set of *predicted* failure values, which makes a negative result falsifiable.

Representations and characters. That a learned transport is a representation, and can be analysed as one, is the premise of Section 7. The tools — character norms, multiplicities, kernels of irreps — are standard [14, 6]. Orthogonal and unitary recurrent networks have a long history [1, 12], and Householder products in particular were introduced there as a parameterisation of the orthogonal group [8, 4]; what is new here is reading the learned map through its character to recover which quotient the model settled on, without access to the group table.

3 Setup

The setup is deliberately not the language model of paper A. There is no input projection and no gate: each group element g owns a learned table of n_h reflection vectors $\mathbf{u}_{g,1..n_h} \in \mathbb{R}^d$ and coefficients $\beta_{g,i}$, and the transport it induces is their product,

$$T_g = \prod_{i=1}^{n_h} (I - \beta_{g,i} \hat{\mathbf{u}}_{g,i} \hat{\mathbf{u}}_{g,i}^\top), \quad \hat{\mathbf{u}} = \mathbf{u} / \|\mathbf{u}\|,$$

initialised at $\beta = 2$, i.e. at pure reflections. The state starts at a learned $\mathbf{h}_0$, each token applies its own T_g , and a linear readout maps the state to the $|G|$ classes. Giving every element its own transport removes the question of whether a shared projection can separate the elements and leaves only the question this paper asks: can a *product of reflections* realise the group?

Training reads a word $g_1 \dots g_L$ and is supervised on the running product at every prefix, with an explicit group-law penalty added to the cross-entropy:

$$\mathcal{L} = \text{CE}(\text{readout}(\mathbf{h}_t), g_1 \dots g_t) + \lambda \mathbb{E}_{\mathbf{h},a,b} \|T_b T_a \mathbf{h} - T_{ab} \mathbf{h}\| / \sqrt{d}, \quad \lambda = 1,$$

the expectation taken over 64 random triples per step. The penalty asks directly for a homomorphism rather than hoping one emerges, which is what makes a failure interpretable: a model that ends with a small penalty and sub-unit accuracy has learned *some* homomorphism, and the question is which.

Unless stated otherwise: $n_h = 4$, $d = 4$ for S_4 and Q_8 and $d = 5$ for A_5 , AdamW at 2×10^{-3} with weight decay 0.01 and gradients clipped at 1.0, batch 256, 2000 steps, trained at length $L = 12$ and evaluated at $L = 12$ and $L = 96$ on 1024 held-out words each. A run counts as *flat*, and so as a candidate homomorphism, when the two accuracies differ by less than 0.05.

4 The abelian ceiling

Before asking which quotient a transport learns, it is worth fixing a bound that no amount of training can move, because it converts one architectural property — whether the transports commute — into a number.

Proposition 1 (multiset bound) *If the transports commute, the accuracy on the length- L word problem of G cannot exceed*

$$C(G, L) = \sum_M \Pr[M] \max_{g \in G} \Pr[g_1 \dots g_L = g \mid M],$$

the sum running over multisets M of L elements.

Proof. If $T_a T_b = T_b T_a$ for all a, b , the state after reading a word depends on the word only through how many times each element occurs, so the prediction is a function of M . The best such function returns, for each M , the most probable product, which is the stated sum. $\square$

The bound is computable, and we evaluate it exactly wherever enumeration is feasible. At $L = 8$ over all 1287 multisets of six elements, $C(\mathbb{Z}_6, 8) = 1.0000$ and $C(S_3, 8) = 0.3850$; over the 6435 multisets of eight, $C(Q_8, 8) = 0.5323$. The $S_3/\mathbb{Z}_6$ pair is matched in order, so a transport that clears 0.385 on S_3 has done something order-sensitive, and nothing about the size of the group can explain it away.

Table 1: Matched-order test at $L = 8$, three seeds each. The ceiling is proven, not fitted. Commuting transports sit on it; the reflection transport of Section 3 passes it on two seeds of three. Group-law violations for the reflection rows are ≤ 0.005 , so those runs are homomorphisms.

transport	commuting?	$\mathbb{Z}_6$	S_3
diagonal rotation	yes	1.000 ± 0.000	0.388 ± 0.006
single Givens	yes	1.000 ± 0.000	0.389 ± 0.003
Givens butterfly	no	1.000 ± 0.000	0.779 ± 0.380
Householder product ($n_h = 4$)	no	1.000 ± 0.000	0.780 ± 0.382
transformer	—	0.166 ± 0.004	0.167 ± 0.003
<i>proven ceiling for any commuting transport</i>		1.000	0.385

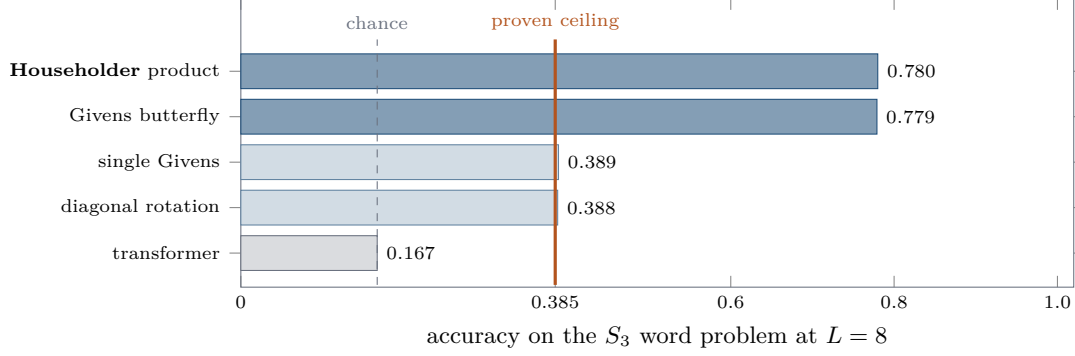


Figure 1: **The ceiling is a theorem, not a baseline.** No commuting transport can exceed 0.385 on S_3 at any width, depth or training budget, because its output can only depend on the multiset of inputs. The two commuting transports (light) sit exactly on it; the two non-commuting ones (dark) pass it. Dashed line: chance, 0.167. The bars are means over three seeds, and the spread behind the two dark bars matters — see the text.

The reflection row is the transport this paper is about, measured here rather than inherited: an earlier ladder in this project reported a non-commuting arm on S_3 at 0.778, but that arm is a Givens butterfly read out through an MLP, not a reflection transport, and it is listed above as such.

The ± 0.382 is the interesting part. It is not noise around a mean: the three seeds return 1.000, 0.339, 1.000. Two solve S_3 outright, and the third sits on $1/3$ — which is a rung of S_3 , whose normal subgroups have orders 1, 3, 6. So the seed that does not clear the ceiling has not failed randomly; it has learned the quotient $S_3/\mathbb{Z}_3 \cong \mathbb{Z}_2$ and is guessing within a coset of size three. That is the phenomenon of Section 5 appearing already in the ceiling experiment, and it explains why the butterfly mean (0.779) and the reflection mean (0.780) agree so closely: both are two solves and one collapse to the same rung, not two measurements of one underlying accuracy.

A second group. Clearing a bound on one group is thin evidence, so we repeat the test on Q_8 , whose ceiling is also exact (0.5323 over 6435 multisets). Three seeds give 0.260, 0.478 and 1.000, with group-law violations ≤ 0.005 throughout: one clears the ceiling outright, and the other two land on rungs of Q_8 ($1/4$ and $1/2$). This is the sharpest place to see that the ceiling and the rungs are different constraints. Q_8 ’s rung at 0.500 sits just *below* its ceiling of 0.5323, so a run that learns the $\mathbb{Z}_2$ quotient stops a hair under a bound it was never subject to — the bound binds commuting transports, and this transport does not commute. What stopped it was the quotient, not the ceiling.

Being non-commuting removes a ceiling; it does not supply an optimiser that reaches above it. The rest of this paper is about where the runs that fall short actually land.

5 The lattice law

A transport that ends training with a small group-law penalty is a homomorphism $\rho : G \rightarrow GL(d)$ composed with a readout — GL rather than O , since β is free and a learned transport need not stay orthogonal. Its kernel $N = \ker \rho$ is a normal subgroup, elements of the same coset of N are indistinguishable to it, and its accuracy is therefore $1/|N|$. The set of achievable accuracies is thus the set $\{1/|N| : N \trianglelefteq G\}$ — the *rungs* — which is a property of the group alone and is invisible to the model. We enumerate the normal subgroups of each group directly (unions of conjugacy classes closed under the product) rather than quoting them: S_4 gives orders 1, 4, 12, 24; Q_8 , being Hamiltonian, gives 1, 2, 4, 4, 4, 8; A_5 , being simple, gives only 1, 60.

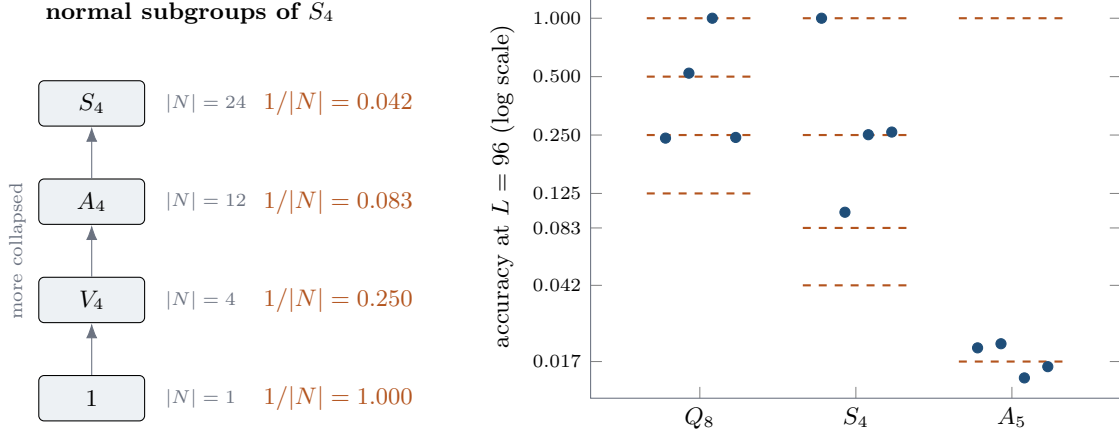


Figure 2: **Failure is quantised onto the normal subgroup lattice.** Left: the normal subgroups of S_4 form a chain, and each one N implies an achievable accuracy $1/|N|$ for a model that has learned G/N . Right: every one of twelve trained runs ($\bullet$) lands on a rung (dashed) of its own group. Q_8 is Hamiltonian, so every subgroup is normal and it has a rung at 0.500 that S_4 does not; A_5 is simple, so it has no rung between 0.017 and 1.000 and nothing may appear there. The model is never shown the lattice.

Table 2: Four seeds per group. Every run is length-flat, and every one lands on a rung of its own group. Worst deviation across all twelve runs: 0.021. Flatness is a necessary condition, not a demonstration of mechanism — that is what Table 3 supplies.

group (rungs)	seed	$L = 12$	$L = 96$	nearest rung (dev.)
Q_8 1.000 0.500 0.250 0.125	0	0.247	0.241	0.250 (0.009)
	1	0.498	0.521	0.500 (0.021)
	2	1.000	1.000	1.000 (0.000)
	3	0.264	0.243	0.250 (0.007)
S_4 1.000 0.250 0.083 0.042	0	1.000	1.000	1.000 (0.000)
	1	0.081	0.100	0.083 (0.017)
	2	0.264	0.251	0.250 (0.001)
	3	0.257	0.259	0.250 (0.009)
A_5 1.000 0.017	0	0.015	0.020	0.017 (0.003)
	1	0.009	0.021	0.017 (0.004)
	2	0.018	0.014	0.017 (0.003)
	3	0.015	0.016	0.017 (0.001)

Which rungs are actually informative. The bottom rung $1/|G|$ coincides with chance, so a run landing there is consistent with the law but tells us nothing: it is the trivial quotient and the absence of learning at once. This applies to all four A_5 runs, at 0.017. The registered falsifier for A_5 — a flat accuracy strictly between 0.017 and 1.000, which a simple group forbids — was not triggered, but neither did any A_5 run solve the task, so the group contributes a consistency check rather than a positive result. The load is carried by the intermediate rungs that are neither chance nor perfect: Q_8 at 0.500 and 0.250, and S_4 at 0.250 and 0.083.

The table also does not record a group-law violation per run, so for its chance-level rows we do not claim a homomorphism was learned at all; the claim is about where runs land, and the mechanism is established separately, on the intermediate S_4 runs, below.

Coset consistency. Accuracy alone could be a numerical coincidence, so we test the mechanism on the errors. If a model learned G/N , its predictions must land in the correct coset of N essentially always, and the kernel is identified as the *smallest* N whose coset score reaches 1.000 — read off the errors, with no reference

to the accuracy. The accuracy predicted by that independently identified N then matches the measured one on all five S_4 runs tested:

Table 3: The subgroup identified by the errors predicts the accuracy. Coset scores are the fraction of predictions landing in the correct coset.

raw acc.	1	V_4	A_4	S_4	identified N	$1/ N $
1.000	1.000	1.000	1.000	1.000	1 (faithful)	1.000
1.000	1.000	1.000	1.000	1.000	1 (faithful)	1.000
0.230	0.230	0.999	0.999	1.000	V_4	0.250
0.081	0.081	0.329	1.000	1.000	A_4	0.083
0.042	0.042	0.165	0.472	1.000	S_4	0.042

The law names a subgroup, not an order. Everything above is consistent with a weaker claim than the one we are making. The value $1/|N|$ depends only on the *order* of N , so a model that merely collapsed by some factor dividing $|G|$ would produce the same table. Separating the two needs a group with several distinct normal subgroups of the same order, and Q_8 is one: being Hamiltonian, every subgroup is normal, and it has three of order four, $\langle i \rangle$, $\langle j \rangle$, $\langle k \rangle$, which share the single rung 0.250. Accuracy cannot tell them apart. The errors can.

The prediction is sharper than “one of them scores 1.000”, because the wrong subgroups have a computable score. If the model learned $Q_8/\langle i \rangle$ it guesses uniformly within a four-element coset of $\langle i \rangle$, and each such coset splits exactly half-and-half across the two cosets of $\langle j \rangle$ — so $\langle j \rangle$ must score 0.5, not chance and not one.

Table 4: Ten seeds on Q_8 ; the three that land on the 0.250 rung as homomorphisms. Exactly one order-4 subgroup is identified in each, the other two sit at the predicted 0.5, and across seeds all three subgroups occur — so the choice is not an artefact of how the group was indexed.

seed	$L = 96$	$\{1\}$	$\langle -1 \rangle$	$\langle i \rangle$	$\langle j \rangle$	$\langle k \rangle$
0	0.234	0.230	0.488	1.000	0.488	0.488
7	0.260	0.219	0.477	0.477	1.000	0.477
9	0.246	0.248	0.514	0.514	0.514	1.000
<i>predicted</i>	0.250	0.250	0.500	1.000 for one, 0.500 for the other two		

The registered falsifier — a rung run in which two or more order-4 subgroups score above 0.95, or none does — was not triggered. What the model learns is a quotient by a particular subgroup, and the accuracy is only the shadow that subgroup’s order casts.

A measurement we withdraw. Our earlier runs also reported a per-run kernel size, obtained by clustering the transports $M[g] = \mathbb{E}_{\mathbf{h}} T_g \mathbf{h}$ and setting $|\ker| = |G|/(\#\text{clusters})$. That estimator is invalid: T_g is linear and $\mathbf{h}$ is zero-mean, so $M[g]$ is a Monte-Carlo zero for every g and the clustering runs on sampling noise. Measured, $\|M[g]\|$ is 5.7% of the typical $\|T_g \mathbf{h}\|$, below the clustering threshold; applied to an *untrained* model whose 24 transports are independent random draws and whose true kernel is trivial, it reports $|\ker| = 8$. It returned orders that are not normal subgroup orders at all — 8 for S_4 , 30 for A_5 — in 4 of 12 runs, which is how the fault was found. Nothing in this paper rests on that column; the kernel is identified by coset consistency instead.

6 The same-order control

The cross-group evidence is confounded by group order, since a 24-element problem and an 8-element problem differ in chance level as well as in lattice. The control removes it: S_4 and $\text{SL}(2, 3)$ both have order 24, and their rungs are disjoint apart from the endpoints.

$$S_4 : 1.000, 0.250, 0.083, 0.042$$

$$\text{SL}(2, 3) : 1.000, 0.500, 0.125, 0.042$$

Across four registered runs (2400 steps, 1200 steps, an S_4 -only run at 1800 steps, and a run at width 3), 28 failures were flat enough to score. Every one landed on a rung of its own group, 13 of them on a rung the other group does not possess, and none within 0.03 of the other group’s exclusive rungs. One run missed its registered tolerance — $SL(2, 3)$ at 0.469 against a predicted 0.500, with coset consistency 1.000 — so the registered verdict is *falsified*, and we report it as such; in binomial standard errors that miss is $z = -2.00$, the only $|z| > 1.96$ among the 28, which says the tolerance was set too tight rather than that the law failed. The runs, their predictions and their verdicts are in the repository.

7 Characters as a table-free diagnostic

Identifying the kernel by coset consistency needs the group table. There is a diagnostic that does not. For a representation ρ with character $\chi(g) = \text{tr } \rho(g)$, the norm

$$\langle \chi, \chi \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g)^2 = \sum_i m_i^2$$

counts irrep multiplicities, and needs only traces of the learned transports — no labels, no Cayley table, no character table. Enumerating every way to build a 4-dimensional representation of S_4 and intersecting the kernels of its parts gives what each value implies:

kernel	quotient realised	attainable $\langle \chi, \chi \rangle$ at $d = 4$
1	S_4 (faithful)	2
V_4	S_3	3, 4, 5
A_4	$\mathbb{Z}_2$	8, 10, 16
S_4	trivial	16

At $d = 4$ the value 2 is attainable *only* by a faithful representation, so it is a sufficient certificate of faithfulness. That statement is conditional on full rank, and the condition is not decorative: at $d = 3$ the value 2 means kernel V_4 , at $d = 2$ kernel A_4 .

The test. A derived table is not evidence, so we test it against trained models. Sixteen seeds are trained on S_4 , and each is read twice, in ways that share no inputs:

- *from the errors* — coset consistency gives the kernel N , as in Section 5. This uses the group table and the model’s mistakes.
- *from the traces* — $\langle \chi, \chi \rangle$ and the multiplicities m_i , by orthogonality. This uses neither labels nor errors, only $\text{tr } T_g$.

If the diagnostic works, the second reading must land in the set the first one permits, on every run. We registered the falsifier before running: a run that is length-flat, has a group-law violation below 0.02, is full rank, and whose $\langle \chi, \chi \rangle$ lies more than 0.25 from every value its coset-identified kernel allows.

Table 5: Sixteen seeds on S_4 . Seven are homomorphisms by the registered tolerance; on all seven the trace reading agrees with the kernel the errors identify, and the multiplicities identify that same kernel a third time. No run triggered the falsifier.

kernel from the errors	runs	allowed $\langle \chi, \chi \rangle$	measured
1 (faithful)	5	2	2.001, 2.002, 2.004, 1.999, 2.000
V_4	2	3, 4, 5	3.001, 2.999
A_4	0	8, 10, 16	—
S_4	0	16	—

The decompositions are exact, not merely close. A faithful run reads $m_{\text{sign}} = 1.000$, $m_{\text{std}} = 1.000$ and the rest zero to three decimals ($1 + 3 = 4$ dimensions, $\langle \chi, \chi \rangle = 2$); a V_4 run reads $m_{\text{triv}} = 0.999$, $m_{\text{sign}} = 1.001$, $m_{2\text{dim}} = 1.001$ ($1 + 1 + 2 = 4$, $\langle \chi, \chi \rangle = 3$), and $\ker = \ker(\text{triv}) \cap \ker(\text{sign}) \cap \ker(2\text{dim}) = V_4$, which is what the errors said.

Integrality is an independent check on the condition. The diagnostic is meaningful only for a transport that *is* a representation, and above we decided that with the group-law penalty — the same quantity the training objective minimises, which is uncomfortably circular. A character is a non-negative *integer* combination of irreps, so the distance of the measured multiplicities from the nearest integer tests the same thing without reference to the objective. The two criteria separate the sixteen runs identically and with room to spare: the seven scored runs have integrality deviation at most 0.005, and the best of the other nine is 0.147, a gap of $29\times$ with no overlap. A run that is not a homomorphism has a $\langle\chi, \chi\rangle$, but it is not a character of anything, and integrality says so without being told.

What this does not cover. No scored run landed on the A_4 or S_4 kernels, so the predicted values 8, 10, 16 remain derived rather than observed — the runs that did reach those kernels ($\langle\chi, \chi\rangle$ of 2.68 and 1.65) had group-law violations of 0.14–0.16 and are not representations. The table is therefore confirmed on two of its four rows. Separately, one run solves the task at accuracy 1.000 while failing the homomorphism test (violation 0.051, $\langle\chi, \chi\rangle = 1.22$, with $m_{\text{std-sign}} = 1.06$ carrying almost all of it): an approximately three-dimensional faithful representation with a fourth dimension dying. Solving the word problem does not require being an exact homomorphism, and the diagnostic correctly declines to label such a run.

Finally, the diagnostic is better at certifying than at training. Adding a penalty that drives the mean squared trace toward 2.0 — no table, no labels, nothing but traces — solved 2 of 6 seeds against 3 of 6 for the unregularised baseline, so the table-free form does not help. A character-*table* penalty, which supplies the target multiplicities and is therefore not table-free, solved 5 of 6. The distinction matters: knowing the target representation helps optimisation, but the trace norm alone, which is all one can measure without the group, does not supply it.

8 The same invariant governs binding

The diagnostic above is a character: $\chi(g) = \text{tr}(g)$, and $\langle\chi, \chi\rangle$ identifies the learned quotient without the group table. The same quantity controls a capability that has nothing to do with group word problems.

For an orthogonal g acting on the sphere, the expected overlap of a random unit vector with its image is the normalised trace,

$$\mathbb{E}[\mathbf{v}^\top g \mathbf{v}] = \text{tr}(g)/d,$$

which is exactly how much an operation *hides* a vector — the property a binding operation needs if a value is to be stored against a key and recovered later. A single Householder reflection has $\text{tr} = d - 2$, the least-hiding non-trivial element of $O(d)$, and a product of k reflections gives overlap $\approx e^{-2k/d}$. Measured at head dimension 64, with weights set by hand and no training, capacity follows the trace exactly: four associations are recalled at 0.453 with $k = 2$ and at 1.000 with $k = 32 \approx d/2$.

So a product of reflections is a tracker at small k and, *as a bind/unbind scheme*, an associative store at $k \approx d/2$; the character is what separates the regimes. A layer that composes group elements perfectly and cannot retain one fact among four is not exhibiting two unrelated limitations; it is at one end of a single curve.

But the recurrence cannot walk along that curve. We tested the obvious consequence in a trained language model, sweeping n_h with everything else fixed, and it failed — backwards. The trace behaves exactly as derived: measured 0.818, 0.611, 0.364, 0.122 at $n_h = 2, 4, 8, 16$ against $(1 - 2/d_h)^{n_h}$ of 0.766, 0.586, 0.344, 0.118. Recall does the opposite of what more capacity would predict, falling monotonically from 0.113 to 0.004 as n_h rises, ending below chance.

The reason is a mismatch between the scheme and the architecture, and it is worth stating plainly because it was implicit in the claim above. The by-construction store binds a key to a value, sums the pairs, and recovers one by applying that key’s reflections *again* — an involution, so the second application unbinds. A recurrence does neither thing: it never applies a query’s inverse at read time, and every subsequent token applies its own transport to the *whole* state, so a stored pair is re-transported by everything that follows rather than sitting untouched until read. Under those conditions more reflections per token are not more capacity, they are a faster scrambler. The model appears to know it: at $n_h = 2$, where it still has the freedom,

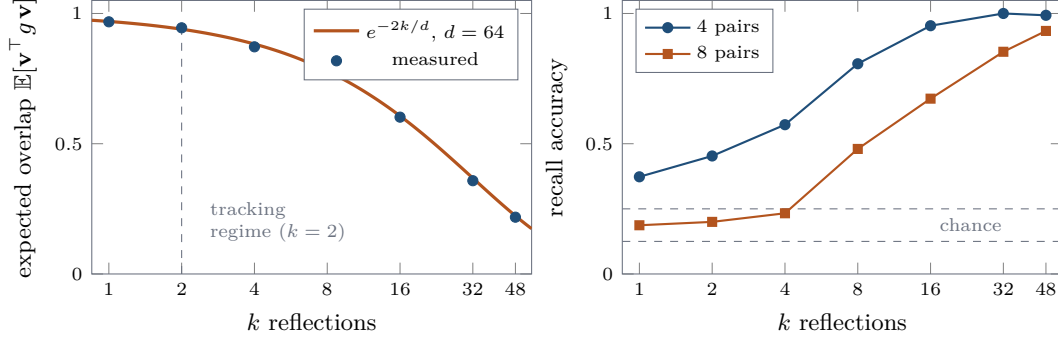


Figure 3: **One knob, two jobs.** Left: the expected overlap of a vector with its image under a product of k random reflections follows $e^{-2k/d}$ to three decimals — the normalised trace, a character. Right: associative recall capacity rises exactly as the overlap falls. A single reflection has $\text{tr} = d - 2$, the least-hiding non-trivial element of $O(d)$, so a transport tuned for state tracking ($k = 2$, dashed) is at the worst possible point for binding, and one tuned for binding ($k \approx d/2$) is too destructive to compose across tokens. No training is involved: the weights are set by hand, $d = 64$.

it settles at $\beta = 1.56$ — away from the pure reflection $\beta = 2$ — keeping the state more nearly intact, while at $n_h = 16$ no β can save it.

The law therefore describes what a product of reflections *could* store. It does not license the claim that raising n_h gives this architecture a memory, and we withdraw that reading. Getting the capacity would need a read path that applies the query’s inverse and some way to shield a stored pair from intervening tokens: a different architecture, not a different n_h .

9 What a quotient is good for

Calling a quotient a failure assumes the task needed the whole group. Change the target to the coset of N and the same collapse becomes the correct answer: the V_4 task asks for the V_4 -coset of the product (6 classes, chance 0.167) and needs only $S_4/V_4 \cong S_3$; the A_4 task asks for the A_4 -coset (2 classes, chance 0.500) and needs only $S_4/A_4 \cong \mathbb{Z}_2$.

Solve rates are 2 of 4 seeds on the full task, 2 of 4 on V_4 and 3 of 4 on A_4 — too few seeds to read a trend from — but the sharper result is where the *failures* land. If the law of Section 5 is really about the group being learned, then on a quotient task it should apply to the quotient, and the rungs should be those of S_3 and $\mathbb{Z}_2$ rather than those of S_4 . The normal subgroups of S_3 have orders 1, 3, 6, giving rungs 1.000, 0.333, 0.167; those of $\mathbb{Z}_2$ give 1.000 and 0.500. Neither 0.333 nor 0.500 is a rung of S_4 .

Table 6: The lattice law one level down. All eight runs land on a rung of the *quotient* they are asked for; worst deviation 0.013.

task	group needed	$L = 96$ accuracy by seed			
V_4 -coset	S_3 (rungs 1.000 0.333 0.167)	1.000	1.000	0.320	0.336
A_4 -coset	$\mathbb{Z}_2$ (rungs 1.000 0.500)	1.000	0.495	1.000	1.000

The failures sit at 0.320 and 0.336 against S_3 ’s rung of $1/3$, and at 0.495 against $\mathbb{Z}_2$ ’s rung of $1/2$. A registered prediction accompanies this: a quotient task does not require a faithful representation, so the amplitude on S_4 ’s 3-dimensional irreps should stay near zero even when the task is solved perfectly. It does — at most 0.042 across all eight quotient runs, against ≈ 1.00 on the solved full-task runs. Capacity tracks the task, not the ambient group.

10 Limitations

Small groups, small dimensions, single-layer transports, four seeds per cell, and a training objective that includes an explicit group-law penalty — the law describes where this architecture lands when it fails, not a claim about trained networks in general. The same-order control is registered as falsified on one run of 28. The Frobenius–Schur account of *which* rung is reached, present in earlier notes, is not claimed here: $SL(2, 3)$ solved the task outright at width 6, which that account does not predict.

Three weaknesses are specific. The bottom rung is chance, so the four A_5 runs confirm the law without demonstrating it, and the positive evidence rests on the intermediate rungs of Q_8 and S_4 and on coset consistency. Clearing the abelian ceiling is shown on S_3 and Q_8 , and on each only by a subset of seeds, so Section 4 bounds what the architecture can express and says little about what training reliably finds. And the character diagnostic is doubly conditional — on the transport being a homomorphism, and on the representation being full rank. Both conditions are now measured rather than assumed, and the diagnostic agrees with the error-based reading on all seven runs that meet them, but two of the table’s four rows (A_4 and S_4 kernels) have no qualifying run behind them and remain derived rather than observed.

Every derived quantity above — the ceiling, the rung sets, the attainable character norms — is recomputed from the group definitions by `evidence/verify_paperB.py`, which also re-reads the measured tables from the result files rather than trusting a transcription.

11 Conclusion

A trained reflection transport that fails a group word problem does not produce an arbitrary number. It produces $1/|N|$ for a normal subgroup N , having learned the quotient G/N — and this is verifiable three ways: the accuracy lands on a rung (worst deviation 0.021 over twelve runs), the *errors* independently identify the same N by coset consistency, and a matched-order control separates the lattice from the group order. The failure is a compression onto an exact algebraic object, not a partial approximation of the right one.

Two consequences seem worth carrying further. First, the collapse is only a failure relative to a task that demands the full group; when the target is itself a quotient, the same behaviour is correct, and the lattice law reappears on the smaller group’s rungs. Real sequence modelling rarely requires distinguishing states with identical continuations, so the set of legal compressions may be a more useful description of a recurrent layer than its accuracy on a task that forbids all of them. Second, the invariant that names the quotient — the normalised trace — is the same quantity that bounds how much the transport can bind and store. Tracking and storage are not two mechanisms in this architecture but two ends of one knob, and a layer that composes group elements perfectly while failing to retain a single fact is exhibiting one property, not two failures.

The main limitation is scale: small groups, small dimensions, four seeds. The law is a statement about where this architecture lands when it fails, and whether the same quantisation governs larger groups, deeper stacks, or transports trained without an explicit group-law penalty is open.

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